

Tool Use · Reasoning · ReAct

Paytm deployed an agent in 2025 that not only answers "what is my refund status?" but CALLS the refund system, updates the user, and emails an acknowledgement. That is the difference autonomy makes.

Which support question in your life should be handled by an agent, not a human?

WHAT'S INSIDE

- 01** What is an agent?
- 02** ReAct pattern
- 03** Designing tools
- 04** Memory — short-term + long-term
- 05** 60-line agent — real code

WHY THIS LESSON MATTERS

An LLM that answers is a chatbot. An LLM that acts is an agent.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Agent = LLM + tools + memory + loop.

2

Implement ReAct (Reason-Act-Observe).

3

Design function-calling tools.

4

Short-term + long-term memory.

5

60-line production agent.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

What is an agent?

Memorise the definition

02

ReAct pattern

Reason → Act → Observe → Repeat

03

Designing tools

What the LLM actually reads

04

Memory — short-term +...

Users expect both

01

PART 1 OF 4

What is an agent?

Memorise the definition

CONCEPT 1 OF 4

What is an agent?

LLM — the reasoner.

Tools — functions the LLM can call.

Memory — short-term conversation + long-term episodic.

Loop — reason → act → observe → decide.

DEFINITION

L + T + M + Loop

Remove any one, you have a chatbot.

DEFINITION

What is an agent?

Memorise the definition

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

ReAct pattern

THOUGHT: what do I need to know?

ACTION: call_tool(name, args).

OBSERVATION: parse tool result.

Repeat until done; then FINAL answer.

LOOP

$R \rightarrow A \rightarrow O$

The canonical agent cycle.

KEY POINTS

- THOUGHT: what do I need to know?
- ACTION: call_tool(name, args)
- OBSERVATION: parse tool result
- Repeat until done; then FINAL answer

Designing tools

Verb-first name: search_web, query_db.

1-2 line description — the LLM picks based on this.

JSON schema for inputs.

Return concise, structured output.

RULE

Name verb; describe clearly

Vague description = wrong tool call.

DEFINITION

Designing tools

What the LLM actually reads

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Memory — short-term + long-term

Short-term: running message history.

Long-term: episodic memory in a vector DB.

Working memory: scratchpad for mid-task notes.

Tool output cache: avoid re-calling expensive tools.

DIFFERENTIATOR

Long-term

"Remember last week" is the user's default assumption.

KEY POINTS

- Short-term: running message history
- Long-term: episodic memory in a vector DB
- Working memory: scratchpad for mid-task notes
- Tool output cache: avoid re-calling expensive tools

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

3 tools, 10 minutes

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

What would happen if your agent loops 20 times on one query?

Q2

Which destructive tool needs human approval?

Q3

Which task would your family pay for an agent to do?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Agent = LLM + ? + ? + loop:

A

Memory + Speed

B

Tools + Memory

C

Models + Users

D

Queries + DB

Agent = *LLM* + ? + ? + *loop*:

CORRECT ANSWER



Tools + Memory

WHY

Tools + memory + reasoning loop

ReAct steps:



Read-Act-Reset



Reason-Act-Observe



Remote-Access-Only



React-Ask-Output

ReAct steps:



CORRECT ANSWER

Reason-Act-Observe

WHY

Reason → Act → Observe → Repeat

Why cap iterations?

A

Style

B

Infinite loops common

C

Speed

D

Memory

Why cap iterations?



CORRECT ANSWER

Infinite loops common

WHY

Agents can loop forever on confused tasks; always cap

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

LLM — the reasoner.

Tools — functions the LLM can call.

Memory — short-term conversation + long-term episodic.

Loop — reason → act → observe → decide.

DEFINITION

L + T + M + Loop

Remove any one, you have a chatbot.

THE TAKEAWAY

"Until the user's question is answered."

WHAT TO NOTICE

1

What is an agent?

2

ReAct pattern

3

Designing tools

4

Memory — short-term +...

Ship an agent

Build a 60-line agent with 3 tools.

DELIVERABLES

- Tools: web_search, calculator, send_email.
- Run ReAct loop; log every step.
- Handle 5 test queries.
- Push to GitHub.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Repo + 5 (query, trace) examples.

ONE THING TO REMEMBER

“

"Until the user's question is answered."

— AI Learning Hub

END OF LESSON 5

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 6 — Fine-tuning with LoRA.